UIU RESCUE ROVER TEAM (#URRT)

URRT-PINMAP-V1.3 • TABLE 7

HARDWARE GPIO PINOUT & INTERFACE SPECIFICATION

URRT Rulebook Table 7 Compliance Mandate:
Dedicated optical isolation or logic level shifting between 5V sensors and 3.3V ESP32 logic.

1. ESP32-WROOM-32 Primary Controller Pinout

GPIO PIN	SIGNAL NAME	TARGET SUBSYSTEM	TYPE	VOLTAGE	TABLE 7 ROLE
GPIO 4	ESTOP_INT	Hardware E-Stop Switch	INTERRUPT	3.3V (Pull-up)	Hard Cutoff / Failsafe (5m)
GPIO 5	SONAR_TRIG	HC-SR04 / JSN-SR04T	OUTPUT	3.3V	10us ultrasonic pulse
GPIO 18	SONAR_ECHO	HC-SR04 / JSN-SR04T	INPUT	3.3V (Divider)	Stop Zone: 80-120mm (15m)
GPIO 14	LEFT_PWM	BTS7960 / L298N	PWM (20kHz)	3.3V / 5V Logic	Left Track Speed (0-255)
GPIO 27	LEFT_DIR	BTS7960 / L298N	OUTPUT	3.3V / 5V Logic	Left Direction Control
GPIO 26	RIGHT_PWM	BTS7960 / L298N	PWM (20kHz)	3.3V / 5V Logic	Right Track Speed (0-255)
GPIO 25	RIGHT_DIR	BTS7960 / L298N	OUTPUT	3.3V / 5V Logic	Right Direction Control
GPIO 21	I2C_SDA	MPU-6050 / BNO055	I2C DATA	3.3V	Fast-Mode 400kHz I2C Bus
GPIO 22	I2C_SCL	MPU-6050 / BNO055	I2C CLOCK	3.3V	Attitude & Tilt Danger
GPIO 34	ENC_LEFT	Optical Wheel Encoder	INTERRUPT	3.3V	Odometry & Distance mm
GPIO 35	ENC_RIGHT	Optical Wheel Encoder	INTERRUPT	3.3V	Odometry & Speed mm/s
GPIO 12	LED_AUTO	Autonomous Status LED	OUTPUT	3.3V	Lit when Auto Lock Active (20m)
GPIO 13	LED_TELEOP	Teleoperation LED	OUTPUT	3.3V	Lit in Teleop Mode (x1)
GPIO 2	AUDIO_DAC	PAM8403 Amplifier	DAC / PWM	3.3V	Two-Way Audio Acuity (15m)

2. Single Board Computer (Raspberry Pi 4 / 5 / Jetson) Vision Rig

PORT / PIN	INTERFACE	CONNECTED DEVICE	TABLE 7 MISSION CRITERION
USB 3.0 (Bus 1)	UVC Video 1080p 30fps	Primary Mast Camera	Live Multi-Cam Stream & QR Decode (10m)
USB 3.0 (Bus 2)	UVC Video 1080p 30fps	Chassis Tool Camera	Colour & Shape Analysis, 5s Motion (15m)
UART (Pin 8/10)	Serial 115200 Baud	ESP32 Core Telemetry Link	<= 100ms Polling Latency Benchmark (10m)
Ethernet / Wi-Fi	TCP / WS Port 5000	Ground Control Cockpit	Zero-Latency Teleoperation & Video WebRTC